

potential portfolios to the vendor so that they can perform the analysis. The vendor will then analyze the basket, estimate costs, and send the results back to the manager.

This approach, however, requires that the information be passed from the manager's site to the vendor's server where the data will be computed, and possibly even stored, before being sent back to the manager. Investors who are using the process need to ensure that their data queries are not being saved or stored at the vendor's or broker's site without their prior approval. If the data is stored on the vendor's site it could potentially allow parties outside the manager's firm to reverse engineer the investment decision.

To alleviate this fear, some vendors provide results of their models to managers in the form of cost curves and include a specified universe of stocks with various sizes and execution strategies. PMs can query and filter these data points for the stocks they are interested in analyzing, but this is a very inefficient process and requires an iterative approach to determine optimal solutions.

Portfolio managers continuously state that they are leery of anything that could potentially result in any kind of information leakage or reverse engineering of the investment decision process. And rightly so! It is the stock selection and portfolio construction process that is the true value of the manager. Even if managers use a verified secure FTP or API protocol that is not viewable by the vendor, the process still does not allow investors to incorporate their own proprietary variables into the analysis. For example, they still cannot integrate proprietary volatility estimates or expected liquidity conditions into the model to perform "what-if" analysis. And we have yet to meet a portfolio manager willing to share their proprietary alpha estimates with any outside party for improved pre-trade analysis.

If the vendor or broker will not provide the model to managers what are they to do? Managers could develop their own market impact model using tick data or by incorporating their own inventory of orders and trades to calibrate the model. But this is very often very time consuming, resource intensive, and could suffer from in-sample error if they rely only on their own trade data. An alternative approach is for managers to develop and build their own models but incorporate broker and vendor pre-trade cost estimates to calibrate the model. This will also allow the managers to incorporate their proprietary views of liquidity, volatility, and even their own alpha estimates.